
Performance of Volatility asset as Hedge for investor's portfolio against Tail Risk Events: COVID19 and 2008 Financial Crisis

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Abstract

Under extreme tail events, most of the asset prices tend to be positively correlated breaking the diversification benefits. Tail risk hedging strategies come to the rescue of portfolio managers in these extreme market movements. The purpose of this paper is to evaluate “Volatility” as an asset class and explore its tail risk hedging performance under the two black swan events i.e. COVID 19 and 2008 Financial Crisis. We construct portfolio using various combinations of asset classes with and without Volatility Index to study the portfolio performance with respect to the two tail risk events. Empirical results indicate that with just allocation of 5% of portfolio to volatility asset class, investors with different risk appetite were able to achieve a 10% expected returns with reduced uncertainty. Moreover, the portfolio performance of Equity, Debt, Commodity and Bitcoins can be matched or outperformed with portfolio of Equity, Debt and Volatility asset class and with lesser number of stocks under each category.

Highlights:

- The study is to explore “Volatility” as an asset class and explore its tail risk hedging performance under the two black swan effects i.e. COVID 19 and 2008 Financial Crisis
- Empirical results show that by holding 5% or 1% of the portfolio allocation in volatility related asset class would have given portfolio insurance against the two Black Swan events.
- Moreover, the portfolio of Equity, Debt, Commodity and Bitcoins performance can be matched or outperformed with portfolio of Equity, Debt and Volatility asset class and with lesser number of stocks under each category

Keywords: portfolio performance, tail risks, volatility Index, efficient frontier

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1: Introduction

The basic tenets of finance with respect to investment management is that market pays you for bearing systematic risk as unsystematic risk can be diversified (Markowitz,1952). Portfolio diversification within and across asset classes and geographies is carried out by fund managers not to maximize returns but limit the impact of volatility on their portfolio to hedge their concentration risk. The strategy of diversification works under the assumption that different asset classes are negatively correlated or independent. Szado (2009) confirmed that as the need for diversification grew during the crises, the ability of many assets to hedge equity holdings fell at the most inopportune time.

Global markets are more interconnected and networked like never before due to globalization, technological advancement and financial innovation. Well diversified portfolios are often susceptible to significant risk associated with a sustained fall in equities prices. Traditionally, investors have invested in fixed income instruments to offer some protection during equity bear markets. It has been observed that equities and bonds show a high negative correlation during times of exceedingly high volatility (Connolly et al. (2005)). Recent events of market downturn (COVID-19, Global financial crises of 2008 etc.) indicate that extreme market moves occur more frequently than those predicted by the statistical models. Investors are risk averse, wish to limit volatility and seek to avoid negatively skewed payoffs at times of extreme market movements (Kumar (2009)). Diversification strategies typically break down at these times and it is a problem for the portfolio managers who on one hand need to generate risk adjusted returns for their investors and on the other hand they need to insulate their portfolio for potential losses. The traditional strategy is to look for higher risk adjusted returns during normal times and at times of stress in the financial markets de-risk your portfolio and accept lower alphas. The undesirable outcome of this strategy is that risky assets are held when returns are lowest and transition to safety occurs when low risk assets are relatively much more expensive.

Some of the extreme tail risk events observed includes Great Depression of 1930, Oil Embargo Arab-Israeli War 1973, 1987 Black Monday, 1997 Asian Financial Crisis,

“Dotcom” Crash, 9/11 Attacks, 2008 Global Financial Crisis, Brexit and the recent COVID19 pandemic. In finance, the term tail risk events typically refer to infrequent, outsized downside market moves (statistically speaking; three standard deviations or more from its current price) such as those observed during the 2008 financial crisis and are seen now during COVID-19. While the market decline can be severe under Black Swan event, most have been short-lived. Markets typically returned to previous peak levels in no more than a couple of months. The Oil Embargo, Black Monday and the 2008 Global Financial Crisis are notable outliers, with the recovery spanning a year or more. After Black Monday, the Federal Reserve reaffirmed its readiness to provide liquidity, and the market recovered in about 400 trading days. Both the 1973 Oil Embargo and 2008 Global Financial Crisis led to U.S. recessions, lengthening the recovery over multiple years. Also the interval at which such Black Swan events occurred had reduced recently with nearly 5 tail risk events in last 20 years. In fact, under recent COVID19, the S&P 500 index reached a bear market in just 16 days along with other developed and developing market indices, the fastest time period on record.

Tail risk hedging strategies come to the rescue of portfolio managers in these extreme market movements. Some market participants use options to try to mitigate tail risks. One may use “collar option strategy” when holding an underlying portfolio, one may sell the out of the money calls and its premium can be used to buy the out of the money puts. This creates a floor and ceiling to potential profit and loss from the movement of the underlying asset. VIX futures provide volatility based hedging strategy. The VIX index tracks the volatility of S&P 500 options and is commonly referred to as the “investor’s fear gauge index” (Whaley (1993, 2000)). It is an aggregation of the implied volatilities of the puts and calls that expire approximately within a month’s time. Equity market sell-offs are typically accompanied by a spike in market volatility. Similar volatility index was developed by other countries like France, Germany, Japan and Hongkong with large option market. India do have its own volatility index, IndiaVIX using Nifty50 options based on modified CBOE VIX methodology to adapt to the Indian market dynamics. Adding VIX futures to portfolios can significantly lead to reduction in portfolio volatility (Dash and Moran (2005), Black (2006) and Moran and Dash (2007)). Dennis et al. (2006), Daigler and Rossi (2006) and DeLisle et al. (2011) document that VIX provides hedge against declines in the S&P 500.

The objective of this study is to consider “Volatility” as an asset class and explore its tail risk hedging performance under the two black swan events i.e. COVID 19 and 2008 Financial Crisis. We evaluate the portfolio performance of portfolio of Equity, Debt, Commodity and

Cryptocurrency with and without volatility asset class. One-year time window of the two events (COVID19 and 2008 Financial Crisis) was considered to gauge the performance of the portfolio. Empirical results indicate that with just allocation of 5% of portfolio to volatility asset class investor with different risk appetite was able to achieve a 10% expected return with reduced uncertainty. Moreover, the portfolio of Equity, Debt, Commodity and Bitcoins performance can be matched or outperformed with portfolio of Equity, Debt and Volatility asset class and with lesser number of stocks under each category. Based on the results of the VIX performance in stabilizing portfolio returns under Black swan events, indicate the effectiveness and importance of holding volatility as an asset class in avoiding market shortfalls by hedging downside risk. The results are in line with Berkowitz & Delisle (2018) and it further opens door for developing derivatives market based on volatility Index (ETFs tracking performance of such index) in emerging countries like India to mitigate downside risk.

The rest of the paper is organized as follows. In section 2, we outline the research design of our study where we discuss the data sources and its description and methodology. In section 3, we chart out the performance of different asset classes including Equity, FX, Bonds, Commodity and Bitcoins over these two tail risk events on a common scale. In section 4, we construct efficient frontier portfolio optimization using various combinations of asset classes with and without Volatility Index to study the portfolio performance with respect to the two tail risk events. We discuss the summary and conclusions in Section 5.

2. Research Design

Data Sources and its Description:

For our analysis, we use two recent extreme tail events corresponding to COVID19 pandemic and 2008 financial crisis. We have taken the entire financial year (1 April 2008 to 31 March 2009 and 1 April 2019 to 31 March 2020) for portfolio construction at the start of the corresponding financial year and tracking its performance over the whole of corresponding financial year. The time series data (refer Exhibit 1 below) extracted are all rescaled to level of 100 USD at the start of each window i.e. 1 April 2008 and 1 April 2019 for homogeneity of comparisons. Assets in non-USD currency are converted to USD equivalent using the corresponding date's closing spot exchange rate.

Exhibit 1: List of Symbol and source

Symbol	Name	Asset Type	Source
ACWI	iShares MSCI ACWI ETF	Equity Index	Yahoofinance
^GSPC	S&P 500	Equity Index	Yahoofinance
^HIS	HANG SENG INDEX	Equity Index	Yahoofinance
^N225	Nikkei 225	Equity Index	Yahoofinance
^NSEI	NIFTY 50	Equity Index	Yahoofinance
^STOXX50E	Euro STOXX 50	Equity Index	Yahoofinance
AAPL	Apple Inc	Stock	Yahoofinance
MSFT	Microsoft Corporation	Stock	Yahoofinance
AMZN	Amazon.com	Stock	Yahoofinance
GOOGL	Alphabet Inc	Stock	Yahoofinance
JPM	JPMorgan Chase & Co	Stock	Yahoofinance
XOM	Exxon Mobil Corporation	Stock	Yahoofinance
JNJ	Johnson & Johnson	Stock	Yahoofinance
RELIANCE.NS	Reliance Industries Limited	Stock	Yahoofinance
TCS.NS	Tata Consultancy Services Limited	Stock	Yahoofinance
INFY.NS	Infosys Limited	Stock	Yahoofinance
HDFCBANK.NS	HDFC Bank Limited	Stock	Yahoofinance
HDFC.NS	Housing Development Finance Corporation Limited	Stock	Yahoofinance
CTXS	Citrix Systems	Stock	Yahoofinance
ZM	Zoom Video Communications	Stock	Yahoofinance
NVDA	NVIDIA Corporation	Stock	Yahoofinance
CODX	Co-Diagnostics, Inc.	Stock	Yahoofinance
MRNA	Moderna, Inc.	Stock	Yahoofinance
ROG	Rogers Corporation	Stock	Yahoofinance
JNJ	Johnson & Johnson	Stock	Yahoofinance
NVS	Novartis AG	Stock	Yahoofinance
MRK	Merck & Co	Stock	Yahoofinance
PFE	Pfizer Inc	Stock	Yahoofinance
^DRG	NYSE ARCA PHARMACEUTICAL INDEX	Pharma Index	Yahoofinance
GLD	SPDR Gold Shares	Gold ETF	Yahoofinance
SLV	iShares Silver Trust	Sliver ETF	Yahoofinance
CL=F	Crude Oil May 20	Crude Future	Yahoofinance
USO	United States Oil	Oil ETF	Yahoofinance
OIL	iPath® S&P GSCI® Crude Oil Total Return Index ETN	Oil ETF	Yahoofinance
TLT	iShares 20+ Year Treasury Bond ETF	Bond ETF	Yahoofinance
SHY	iShares 1-3 Year Treasury Bond ETF	Bond ETF	Yahoofinance
SHV	iShares Short Treasury Bond ETF	Bond ETF	Yahoofinance
IEF	iShares 7-10 Year Treasury Bond ETF	Bond ETF	Yahoofinance
BWX	SPDR Blmbg Barclays Intl Trs Bd ETF	Bond ETF	Yahoofinance
AGG	iShares Core US Aggregate Bond ETF	Bond ETF	Yahoofinance
BND	Vanguard Total Bond Market ETF	Bond ETF	Yahoofinance
LQD	iShares iBoxx \$ Invt Grade Corp Bd ETF	Bond ETF	Yahoofinance
BSV	Vanguard Short-Term Bond ETF	Bond ETF	Yahoofinance

TIP	iShares TIPS Bond ETF	Bond ETF	Yahoofinance
MBB	iShares MBS ETF	Bond ETF	Yahoofinance
FBNDX	Fidelity Investment Grade Bond	Bond ETF	Yahoofinance
BTCUSD=X	Bitcoin	Crypto Coin	Yahoofinance
ETHUSD=X	Ethereum	Crypto Coin	Yahoofinance
IndiaVIX	Volatility index based on Nifty options	Volatility Index	Investing.com
VCAC	Volatility index based on CAC options	Volatility Index	Investing.com
VDAX	Volatility index based on DAX options	Volatility Index	Investing.com
VHSI	Volatility index based on HSI options	Volatility Index	Investing.com
VIX	Volatility index based on SPX 500 options	Volatility Index	Investing.com
VNikkei	Volatility index based on Nikkei options	Volatility Index	Investing.com
	TEDRATE	Credit Spread	fred.stlouisfed.org
	LIBOR3M – EFFR	Credit Spread	fred.stlouisfed.org
	ICE BOFA US High Yield Index OAS	Credit Spread	fred.stlouisfed.org
	ICE BOFA CCC & Lower US High Yield Index OAS	Credit Spread	fred.stlouisfed.org

Notes: TEDRATE: calculated as the spread between 3-Month LIBOR based on US dollars and 3-Month Treasury Bill

EFFR: The federal funds market consists of domestic unsecured borrowings in U.S. dollars by depository institutions from other depository institutions and certain other entities, primarily government-sponsored enterprises. The effective federal funds rate (EFFR) is calculated as a volume-weighted median of overnight federal funds transactions reported in the FR 2420 Report of Selected Money Market Rates.

ICE BOFA US High Yield Index OAS : The ICE BofAML Option-Adjusted Spreads (OASs) are the calculated spreads between a computed OAS index of all bonds in a given rating category and a spot Treasury curve. An OAS index is constructed using each constituent bond OAS, weighted by market capitalization. The ICE BofAML High Yield Master II OAS uses an index of bonds that are below investment grade (those rated BB or below).

ICE BOFA CCC & Lower US High Yield Index OAS : represents the Option-Adjusted Spread (OAS) of the ICE BofAML US Corporate C Index, a subset of the ICE BofAML US High Yield Master II Index tracking the performance of US dollar denominated below investment grade rated corporate debt publicly issued in the US domestic market. This subset includes all securities with a given investment grade rating CCC or below. issued in the US domestic market. This subset includes all securities with a given investment grade rating CCC or below.

Methodology

The mean-variance optimization criterion, which was first proposed by Markowitz, corresponds to finding the asset allocation ‘w’ that maximizes the expected returns of the portfolio subject to a penalty related to the variance of portfolio returns and other constraints of investing all asset in a combination of the n assets

Our objective function is the portfolio variance, and we will minimize it with respects to the portfolio weights along with meeting the constraint.

Instead of using the portfolio variance, we will use a little trick and scale it down by a factor of 1/2 to ease our calculations. Since the factor is positive, it does not affect the value of the optimal vector of weights w^* :

$$\min_w \frac{1}{2} \sigma_\pi^2 = \frac{1}{2} W^{-1} \Sigma W$$

$W \rightarrow$ Vector of weights of asset in the portfolio

$\Sigma \rightarrow$ Covariance matrix

One constraint is that portfolio return must be equal to a prespecified level m . In mathematical terms:

$$\mu_\pi = W^{-1} \mu = m$$

$\mu \rightarrow$ Vector of expected returns of asset

Second constraint on the weights called the ‘budget equation’. The sum of all the weights must necessarily equal 1. Our wealth must be entirely invested in a combination of the n assets,

$$W^{-1} \mathbf{1} = 1$$

where $\mathbf{1}$ is a n -element unit vector: $\begin{bmatrix} 1 \\ \cdot \\ \cdot \\ 1 \end{bmatrix}$

Summing it all, we formulate the portfolio selection problem as:

$$\min_w \frac{1}{2} W^{-1} \Sigma W$$

Subject to:

$$W^{-1} \mu = m$$

$$W^{-1} \mathbf{1} = 1$$

This problem is an optimization with equality constraints. We can solve it using the method of Lagrange.

We form the Lagrange function with two Lagrange multipliers λ and γ :

$$L(W, \lambda, \gamma) = \min_w \frac{1}{2} W^{-1} \Sigma W + \lambda(m - W^{-1} \mu) + \gamma(1 - W^{-1} \mathbf{1}) \text{ ----- Eqn. 1}$$

we solve for the first order condition by taking the derivative with respect to the vector w :

$$\frac{\partial}{\partial W} L(W, \lambda, \gamma) = \Sigma W - \lambda \mu - \gamma \mathbf{1}$$

$$\mathbf{W}^* = \Sigma^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1}) \text{ ----- Eqn. 2}$$

We have to find the values for λ and γ and then substitute in the constrain and solving we get the below relation:

we define the following scalars:

$$A = \mathbf{1}^{-1} \Sigma^{-1} \mathbf{1}$$

$$B = \mathbf{1}^{-1} \Sigma^{-1} \boldsymbol{\mu}$$

$$C = \boldsymbol{\mu}^{-1} \Sigma^{-1} \boldsymbol{\mu}$$

$$\lambda = \frac{Am - B}{AC - B^2}$$

$$\gamma = \frac{C - Bm}{AC - B^2}$$

Substituting in equation “2” we get the optimal weight for the return “m”.

The weights of a portfolio depend on the return objective m specified in the constraints. By letting m span the real line, we will obtain the minimum-variance frontier.

We can express the variance of a portfolio located on the boundary of the opportunity set as a function of m:

$$\sigma_{\pi}^2(m) = \frac{Am^2 - 2Bm + C}{AC - B^2}$$

The minimum variance portfolio is the portfolio solving the unconstrained problem:

$$\min_w \sigma_{\pi}^2(m)$$

Solving for the first order condition, we would get the return corresponding to the global variance:

$$\mathbf{m}_g = \frac{B}{A} \text{ ----- Eqn. 3}$$

Now that we have $\mathbf{m}_g$, we can easily get the vector of weights w_g corresponding to global minimum variance portfolio.

$$w_g = \frac{\Sigma^{-1} \mathbf{1}}{A}$$

And the global minimum variance:

$$\sigma_g^2 = \frac{1}{A} \text{ ----- Eqn. 4}$$

Now from the equation “3” and “4” we get the returns and volatility corresponding to the global minimum variance portfolio.

Now spanning the return starting from m_g on the real line we can construct the efficient frontier.

Monte Carlo Method:

Above, we derived mathematical way of deriving the efficient frontier. Another brute force approach is to start with some random portfolio weights and run “N” number of simulations by randomly assigning the weights to each asset in the portfolio and meeting the budget constraint. Under each simulation, we compute the portfolio returns and variance and we plot the results in the return-variance plot. As the number of simulations increases it converges to the efficient frontier derived using the theoretical approach.

In our analysis, we construct efficient frontier using the theoretical approach and Monte Carlo method using 5000 simulation.

Sharpe Ratio:

It is a key measure of risk-adjusted return representing the excess return (over the risk free rate) per unit of total risk taken. The higher the Sharpe ratio of a portfolio, the more efficient the portfolio is and for our analysis we consider risk free rate of 0.5% this was a fair assumption given we are denominating all asset in USD and 10-year US treasury rate is ranging at this level.

Portfolio Performance Measure:

As investor may have different risk appetite hence, we don’t allocate any weights to each asset class rather we pass the entire asset and allow the portfolio optimization function to determine the optimal portfolio under the below set of scenarios to gauge the performance of different combination of asset class.

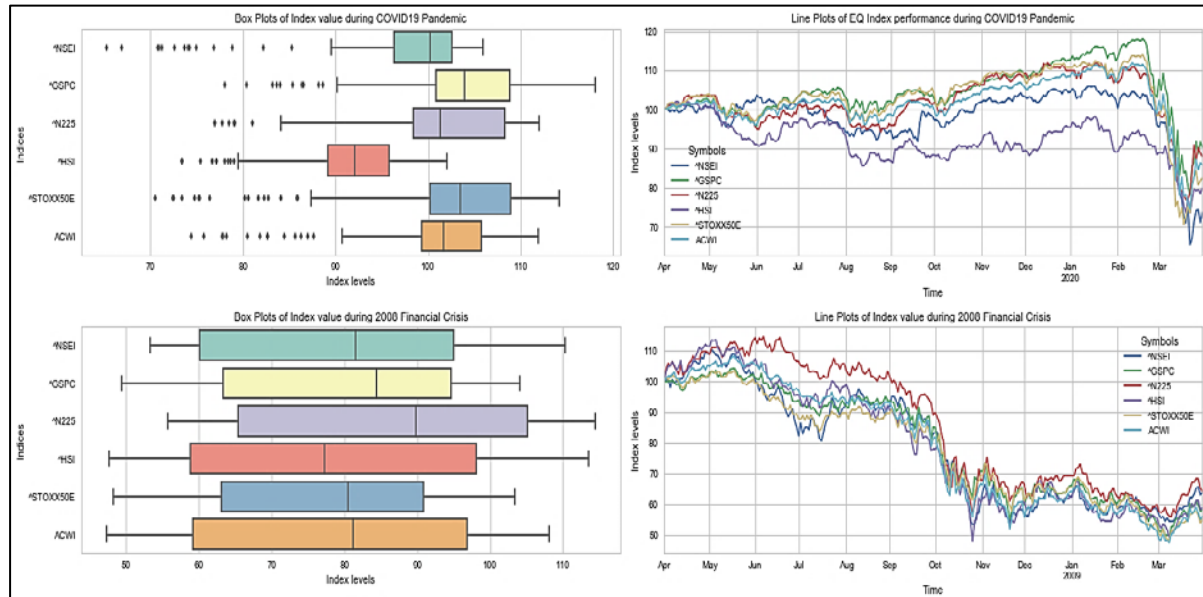
- Maximize the Sharpe Ratio
- Minimize the volatility (Global minimum variance portfolio)
- Achieving a target return of 10% with minimum volatility
- Achieving target volatility of 20% with maximum Sharpe Ratio
-

3. Performance of Different Asset Classes during Tail Risk Events

Equity Market:

The performance of the leading global benchmark indices across Asia, Japan, Europe and America are compared.

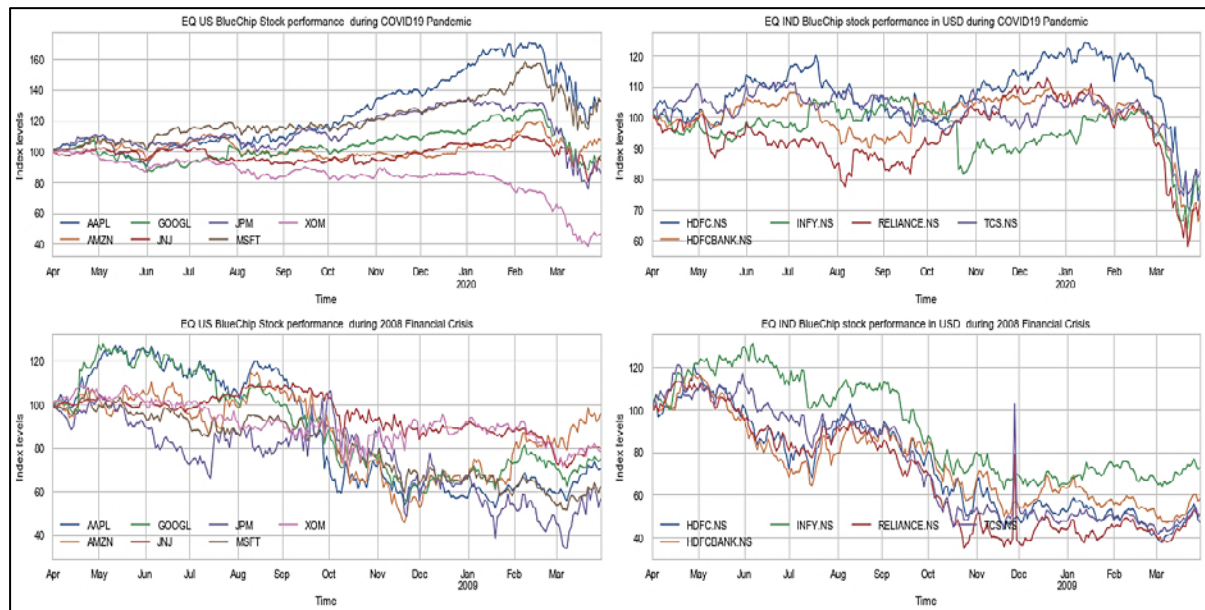
Exhibit 2: Performance of major Equity Indices



It is observed (refer Exhibit 2) that during the financial year 2019-20 all the major global equity indices had dropped by 10-30% and the severity is less than those observed during financial crisis 2008-09. A significant jump is seen in the equity market which was not seen during the financial crisis period. Also from the box plot (refer Exhibit 2), the data clearly shows the large jumps observed on the downside and similar trend is not observed in the box plot of financial crisis. The box and the line plots clearly indicate that the equity markets were trading at generally higher levels with low variations in 2020 than in 2008. The systemic risk of COVID-19 had a global spill over effect directly to the real economy rather than first to the financial sector as was during the time of financial crises of 2008. The policy interventions to arrest the pandemic in 2020 led to “sudden stop” to real activity rather than via financial amplifiers (short-term debt rollover, increase in financing costs to the real economy, leveraged positions etc.). COVID-19 is unique in the sense it is a supply, demand and market shock wherein a downward spiral in the real activity has led to huge financial sector fallouts.

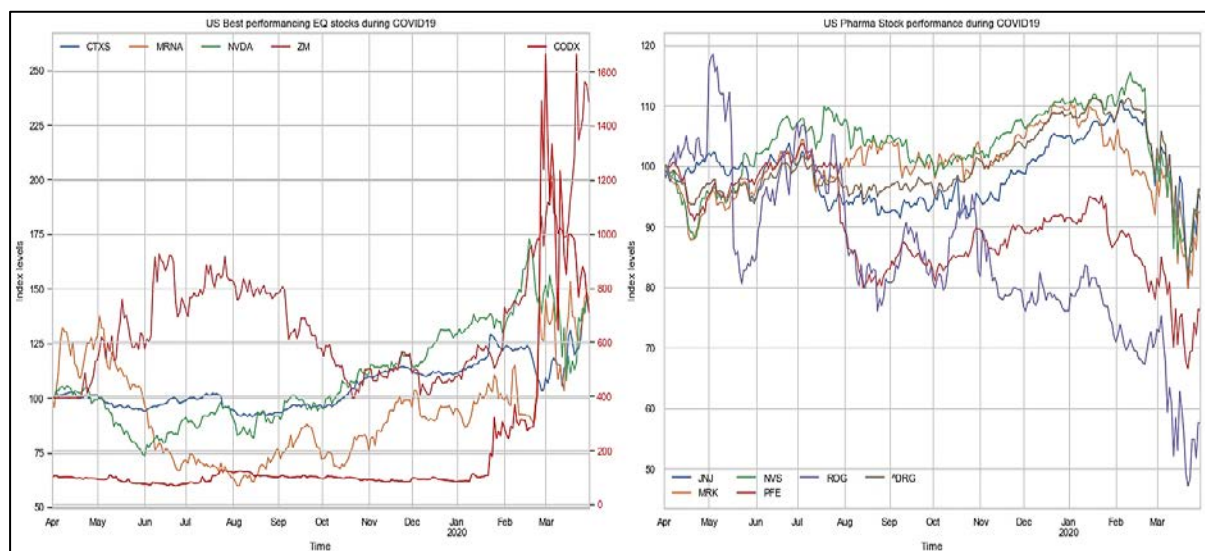
Performance of Blue chip equity stocks:

Exhibit 3: Performance of Blue chip stocks in US and India



The line plots (refer Exhibit 3) clearly indicate that both for USA and India Blue chip stocks, the trend was largely positive before the COVID-19 crises viz-a-viz in 2008 but COVID-19 pandemic event saw a pervasive and precipitous collapse for these stocks.

Exhibit 4: Best performance EQ Stocks during COVID19 and Pharma stocks



Given the COVID-19 crises, the stupendous performance of the Pharmaceutical sector companies on the left pane (refer Exhibit 4) reflects idiosyncratic performance more so as they are the new economy stocks in sectors which are seeing increased interests after the onset of the pandemic. The policy interventions to contain the pandemic includes social distancing and isolation pushing businesses to adapt to remote working thereby popularizing

players like ZOOM, NVIDIA and Citrix systems in virtual private networking, videoconferencing, gaming etc. amongst customers. Further speculations were built on specific Pharmaceutical stocks like Co-Diagnostic, Moderna on the research related COVID19 vaccination. But overall big pharmaceutical companies like Pfizer Corporation, Johnson & Johnson, Novartis and Merck including Pharmaceutical NYSE Pharma index were trading low.

Credit indicator:

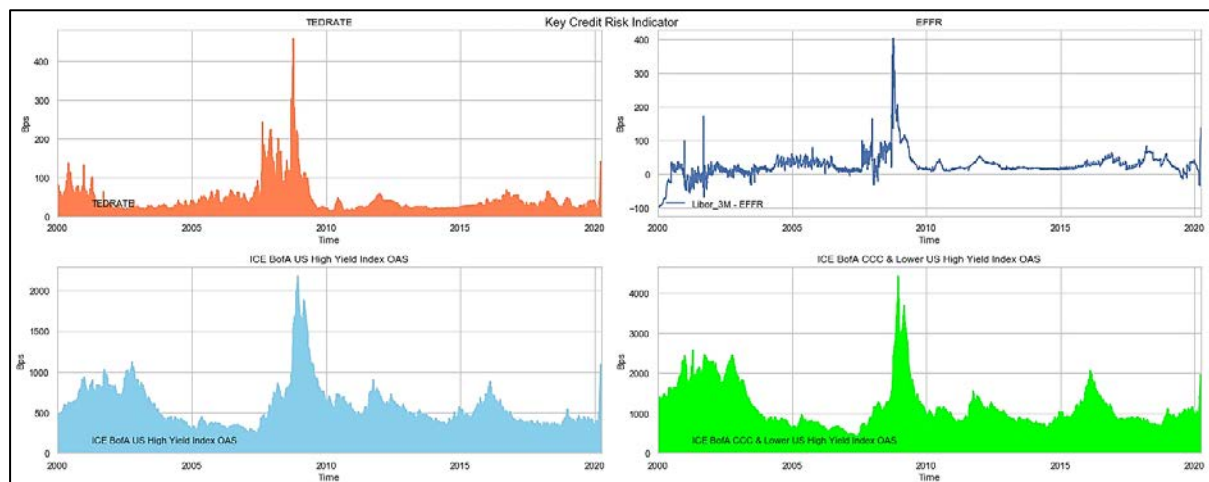


Exhibit 5: Key Credit Spread indicator

The TED spread which is the indication of credit risk can be understood as the difference between the interest rate that investors demand from the government for investing in short term treasuries and the interest rate that investors charge large banks. TED spread often widens in periods of economic crises like it did in 2008 (refer Exhibit 5) which was episode of liquidity crises and is beginning to peak during COVID-19 in response to sudden stop in economic activity. US Fed adjusts its monetary policy stance primarily by changing its target range for the effective federal funds rate (EFRF). EFRF has been predominantly stable except during the global financial crises of 2008 (refer Exhibit 5). The OAS spreads in the credit markets for both the segment (high grade and low grade bonds) peaked in 2008 indicating the higher risk premium as high grade spreads richen. This peak is relatively being scaled again during COVID-19 (refer Exhibit 5).

Bond Market:

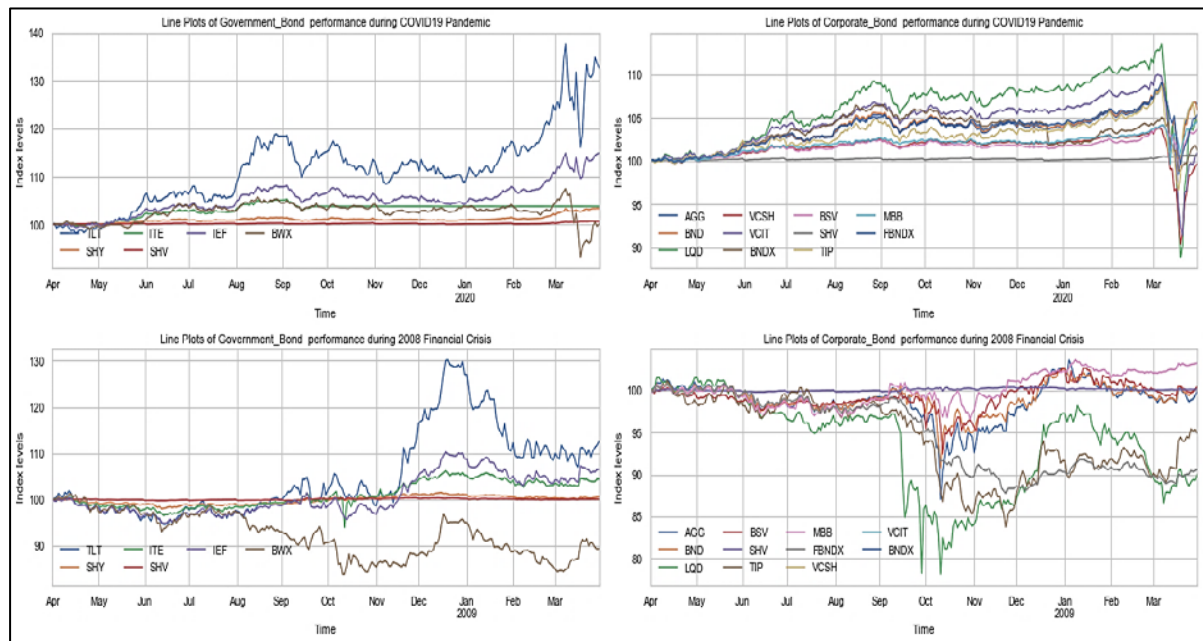


Exhibit 6: Performance of Government and Corporate Bond ETF

The bond markets comprising GSecs of different maturity spectrum are attractive investment destinations at times of global uncertainty and risk off “flight to quality” towards US long term bonds. The rally in the bond market (refer Exhibit 6) before the COVID-19 was driven by a wave of easing Central bank policies both at the USA Fed and around the globe most notably by ECB and BOJ. The demand for the corporate bonds was also robust just before the COVID-19 (refer Exhibit 6) owing to decent macroeconomic conditions, very supportive monetary policy position and supportive demand from non- USA business investors. The scenario was quite different in 2008 (refer Exhibit 6) where the bond markets were not doing well even before the 2008 crises and subsequently post the Lehman crash, the “flight to quality” again led to an increased activity in the GSecs. On the other hand, the corporate bonds suffered heavily with the onset of both the economic crises of 2008 and pandemic of 2020 which can be seen from a steep decline in the graph.

US Interest Rate:

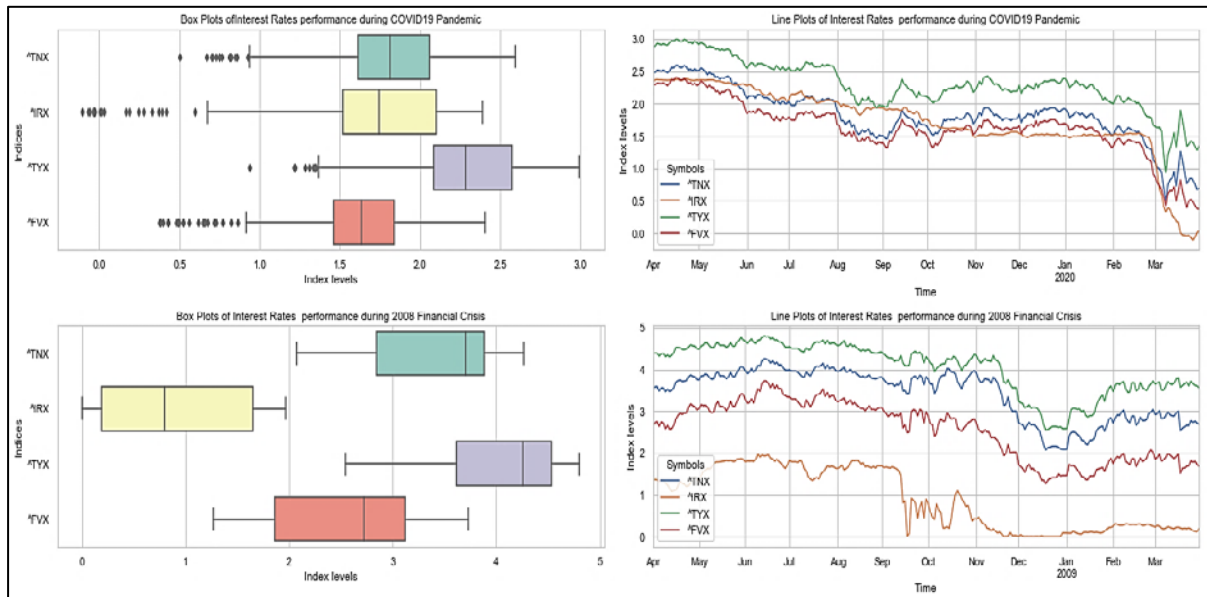


Exhibit 7: Key Interest evolution over Financial Crisis and Covid19

Monetary space was higher for FED in 2008 than in 2019-20 as the base levels prior to the crises in 2008 was higher to support the FED's accommodative stance to support the economy. This is also evident from the box plot diagram of interest rates (refer Exhibit 7) during the two crises periods. The sudden decline in the interest rates at the onset of both the crises shows the immediate action taken by the US Fed to provide monetary stimulus to reinvigorate the economy.

Commodity:

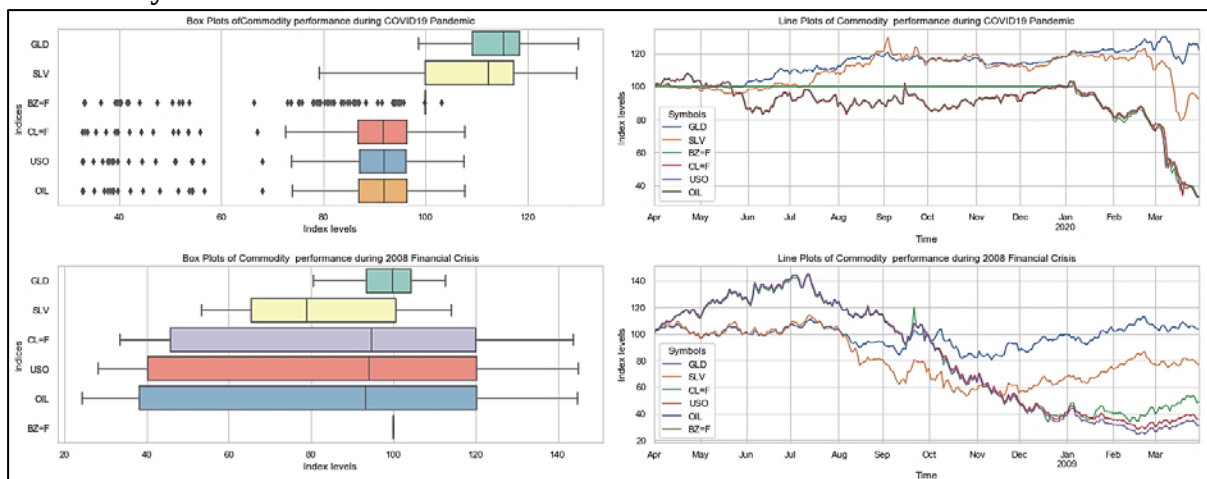


Exhibit 8: Performance of commodity ETF over Financial Crisis and Covid19

Softening of commodity prices during the COVID-19 and global financial crises of 2008 (refer Exhibit 8) were predominantly due to tepid industrial demand due to stifled economic activity.

Bitcoin:

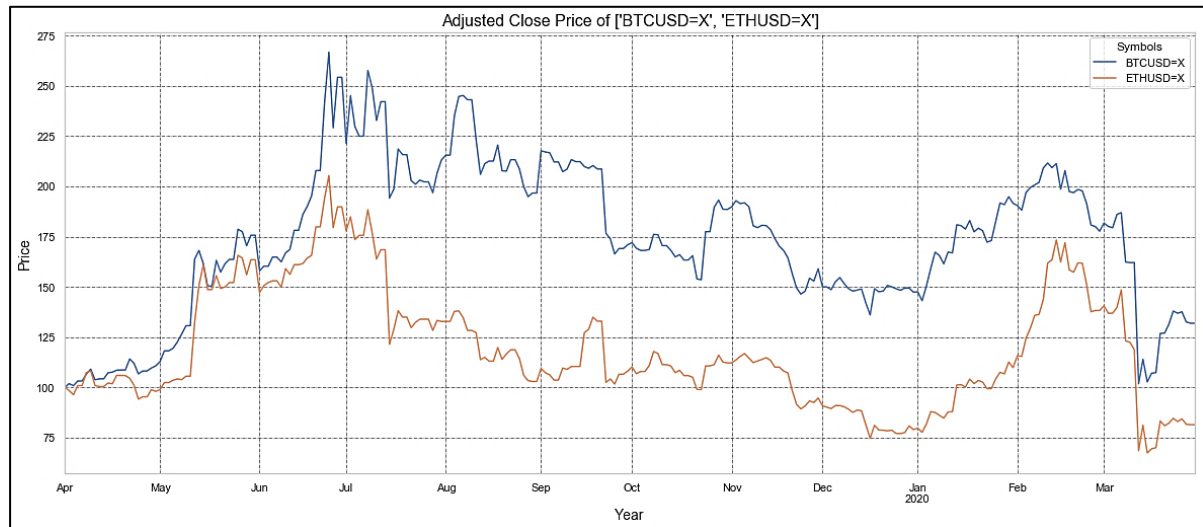


Exhibit 9: Performance of cryptocurrency over Financial Crisis and Covid19

Cryptocurrency like Bitcoin and Ethereum which are considered safe haven assets similar to Gold have also tanked (refer Exhibit 9) following the COVID-19 crises. Traders liquidated their holdings of these Cryptocurrency in the face of pandemic and market turmoil.

Volatility Index:

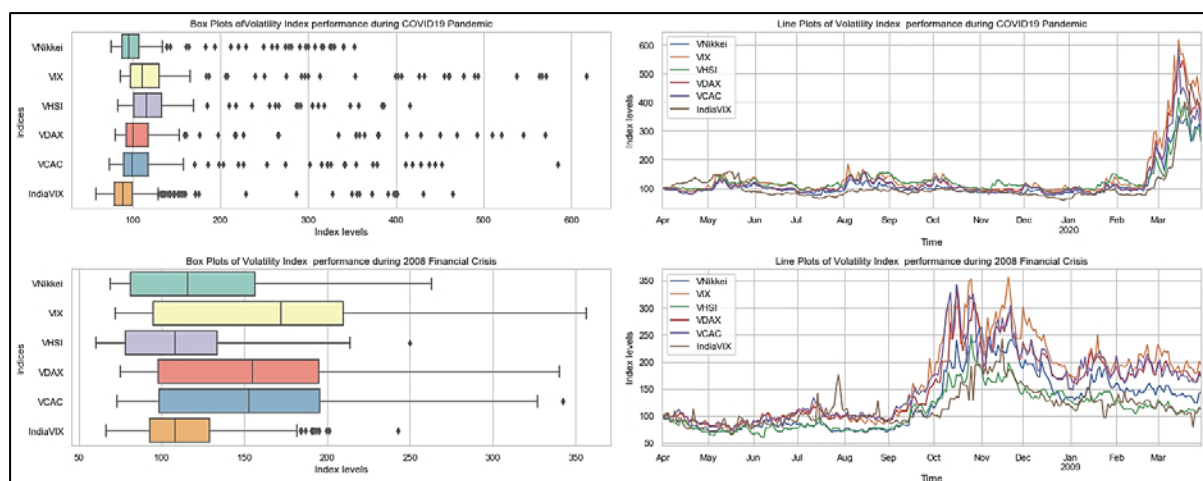


Exhibit 10: Performance of major Volatility indices over Financial Crisis and Covid19

Volatility Indices for all major countries spiked very high (refer Exhibit 10) during the pandemic of 2020 particularly in the month of March 2020 than the levels it reached during 2008 crises. It clearly shows the uncertainty is being weighed very heavily during COVID-19 than it was in during Economic crises of 2008.

4. Efficient Frontier Portfolio Optimization: Performance Analysis

The current low interest environment and expensive equities markets are increasing the pressure to identify sources of return in alternative instruments beyond the conventional asset

classes. Against this backdrop, investors off late are turning to alternative, highly liquid risk premiums. Investors usually use volatility primarily as a source of risk. In fact, in recent years, volatility has become a new asset class for investors looking to diversify their portfolio strategy offering attractive attributes.

We do asset class wise portfolio analysis with and without Volatility asset in the portfolio. Also, we allow the function to decide the weights of each component by maximizing return or Sharpe Ratio or minimizing volatility.

Portfolio of Blue chip EQ stocks from US and India:

Exhibit 11A: Efficient Frontier of Equity stocks during COVID19

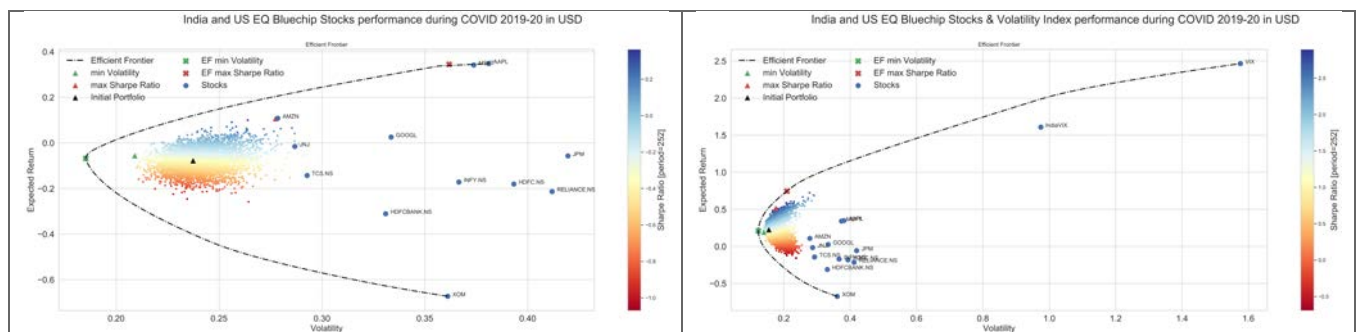


Exhibit 11B: Efficient Frontier of Equity stocks during 2008 Financial Crisis

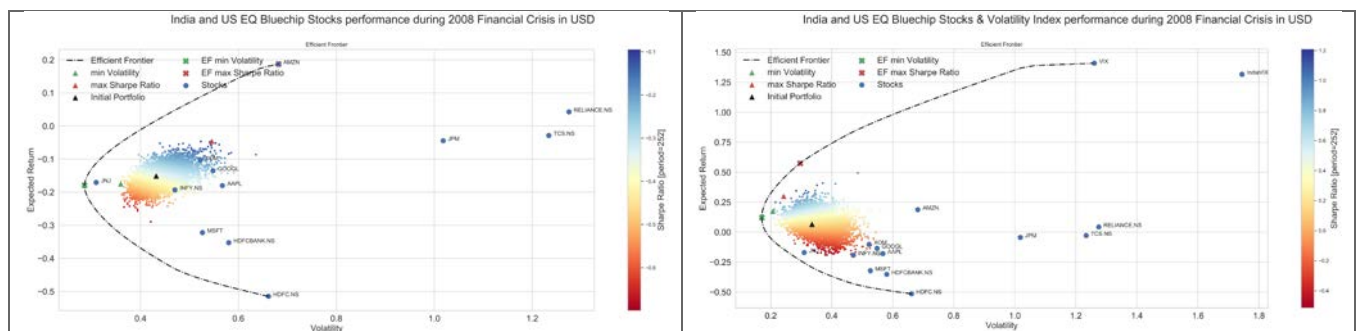


Exhibit 12: Portfolio analysis of Equity stock and Volatility index

Portfolio of Bluechip Equity stock from US and India				
Scenario	COVID 19 FY 2019-20		Financial Crisis FY 2008-09	
Performance Measure	Expected Return %	Volatility %	Expected Return %	Volatility %
20.0% Target Volatility	1.41	20	-17.91	28.55
10.0% Target Return	10	22.92	10	53.84
Maximum Sharpe	34.48	36.18	18.7	68.19
Minimum Volatility	-6.78	18.52	-17.91	28.55
Portfolio of Bluechip Equity stock from US and India and Volatility index				
20.0% Target Volatility	71.19	20	32.44	20
10.0% Target Return	10	12.76	10	17.08
Maximum Sharpe	74.62	20.95	57.4	29.61
Minimum Volatility	20.99	12.31	12.1	17.05

Equity investment would be suitable for investors with higher risk appetite. For such investors with target volatility of 20%, the equity portfolio gave return of 1.41% and -18% under COVID19 and Financial Crisis event respectively (refer Exhibit 12). However, by addition of VIX and IndiaVIX, the return improved to 71% and 32% under COVID19 and Financial Crisis event respectively. Minimum variance portfolio gave negative returns under both the tail events with the financial crisis being more severe. But with the asset allocation of 8% and 4% in VIX and IndiaVIX respectively¹, the minimum variance portfolio return improved to 21% along with reduced volatility of 12% under the COVID19 event. Similar improvement observed during the financial crisis with return of 12% with minimum volatility of 17% by allocation of 17% and 1.4% in VIX and IndiaVIX. The efficient frontier of the portfolio is charted in Exhibit 11.

Portfolio of Blue chip EQ stocks from US and India and US Bond ETF:

Exhibit 13A: Efficient Frontier of Equity stocks and Debt ETF during COVID19

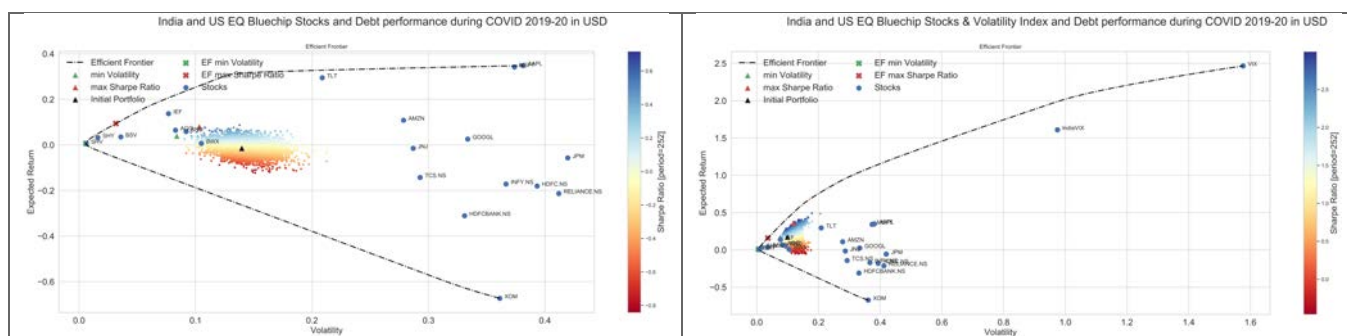


Exhibit 13B: Efficient Frontier of Equity stocks and Debt ETF during 2008 Financial Crisis

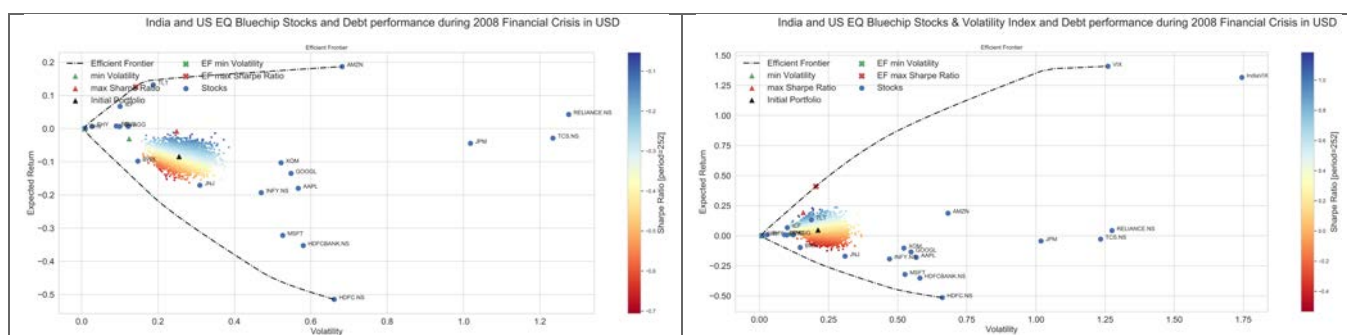


Exhibit 14: Portfolio analysis of Equity stock, Debt ETF and Volatility index

Portfolio of Bluechip Equity stock from US and India and US Bond ETF				
Scenario	COVID 19 FY 2019-20		Financial Crisis FY 2008-09	
Performance Measure	Expected Return %	Volatility %	Expected Return %	Volatility %
20.0% Target Volatility	32.68	20	14.79	20
10.0% Target Return	10	3.39	10	11.23
Maximum Sharpe	9.39	3.16	12.55	14.14
Minimum Volatility	0.64	0.59	-0.07	0.84

Portfolio of Bluechip Equity stock from US and India, US Bond ETF and Volatility index				
20.0% Target Volatility	75.42	20	40.17	20
10.0% Target Return	10	2.23	10	5.02
Maximum Sharpe	15.86	3.58	41.01	20.42
Minimum Volatility	0.87	0.58	-0.01	0.84

Investors with moderate risk appetite would be diversifying between Equity and Debt asset classes and would be expecting some fixed return with low uncertainty. Results (refer Exhibit 14) point out that 10% return can be achieved by allocating the portfolio between Blue chip Equity stocks of India & USA and US Debt market with volatility of 3.4% and 11.2% under COVID19 and Financial Crisis event respectively. But by addition of VIX and IndiaVIX in the portfolio¹, 10% return can be achieved with lower volatility of 2.2% and 5% respectively. The efficient frontier of the portfolio is charted in Exhibit 13.

Portfolio of Blue chip EQ stocks from US and India, US Debt ETF and Commodity:

Exhibit 15A: Efficient Frontier of Equity stocks, Debt and Commodity ETF during COVID19

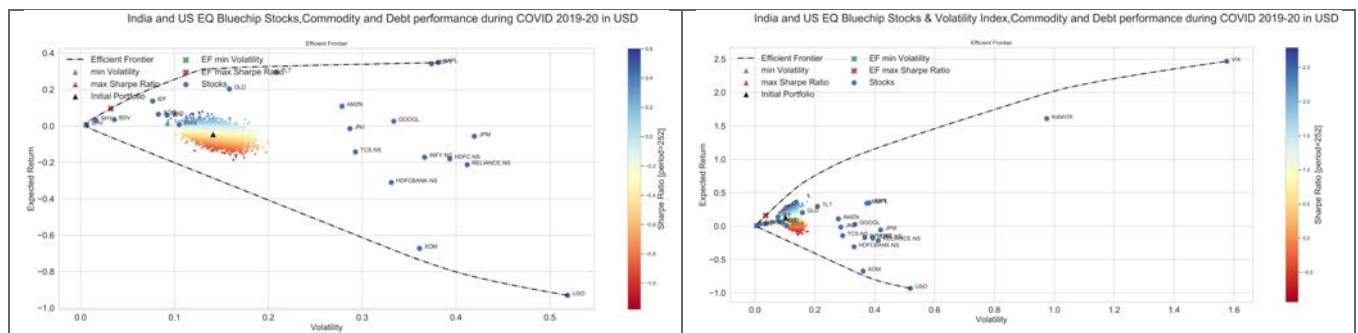


Exhibit 15B: Efficient Frontier of Equity stocks, Debt and Commodity ETF during 2008 Financial Crisis

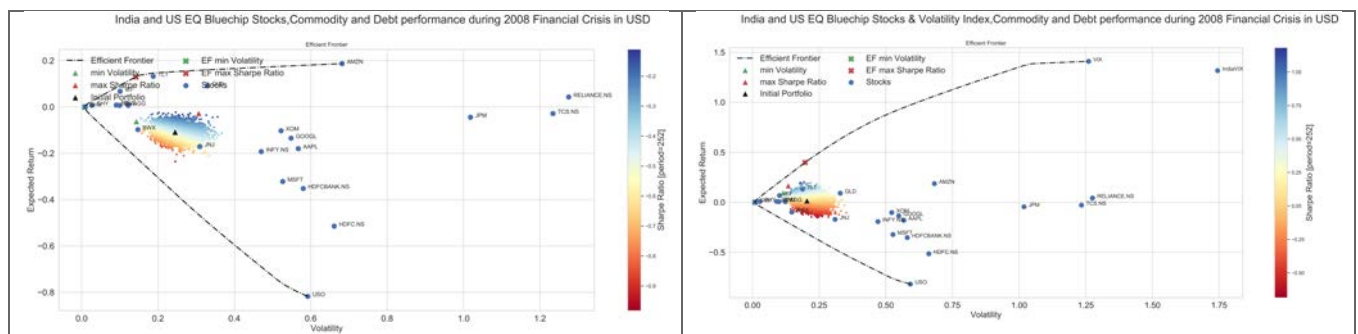


Exhibit 16: Portfolio analysis of Equity stock, Debt ETF, commodity ETF and Volatility index

Portfolio of Bluechip Equity stock from US and India, Bond ETF and Commodity				
Scenario	COVID 19 FY 2019-20		Financial Crisis FY 2008-09	
Performance Measure	Expected Return %	Volatility %	Expected Return %	Volatility %
20.0% Target Volatility	32.68	20	14.79	20
10.0% Target Return	10	3.37	10	10.9
Maximum Sharpe	9.55	3.21	12.96	14.18

Minimum Volatility	0.54	0.59	-0.11	0.84
Portfolio of Bluechip Equity stock from US and India, Bond ETF, Commodity and Volatility index				
20.0% Target Volatility	75.42	20	40.35	20
10.0% Target Return	10	2.23	10	5
Maximum Sharpe	15.85	3.58	39.73	19.69
Minimum Volatility	0.78	0.58	-0.03	0.84

Risk averse Investor would be diversifying risk by investing in wide range of asset classes including Equity, Debt and Commodity like GOLD and OIL and would be expecting some fixed return with very less uncertainty. We observe 10% return can be achieved by allocating the portfolio between Blue chip Equity stocks of India and US, US Debt market and Commodity with volatility of 3.4% and 11% under COVID19 and Financial Crisis event respectively (refer Exhibit 16). But by addition of VIX and IndiaVIX in the portfolio¹, the 10% returns can be achieved with lower volatility of 2.2% and 5% respectively. The performance of the portfolio of Equity, Debt, commodity and Volatility asset class is almost same as portfolio of Equity, Debt and Volatility asset class in achieving 10% returns. This means the diversification benefit derived by inclusion of Gold can be produced by Volatility asset class. Hence, risk averse investors can think of Volatility asset as better hedge or alternate for commodity asset against any Black Swan events. The efficient frontier of the portfolio is charted in Exhibit 15.

Portfolio of Blue chip EQ stocks from US & India, US Debt ETF Commodity and Bitcoin

Exhibit 17: Efficient Frontier of Equity stocks, Debt, Commodity ETF and Bitcoin during COVID19

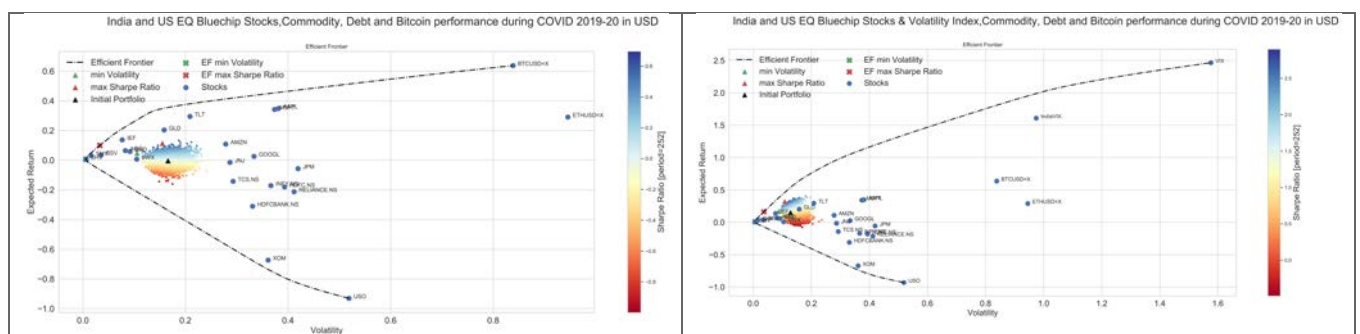


Exhibit 18: Portfolio analysis of Equity stock, Debt ETF, commodity ETF, Bitcoin and Volatility index

Portfolio of Bluechip Equity stock from US and India, Bond, Commodity and Bitcoin				
Scenario	COVID 19 FY 2019-20		Financial Crisis FY 2008-09	
Performance Measure	Expected Return %	Volatility %	Expected Return %	Volatility %
20.0% Target Volatility	37.5	20		
10.0% Target Return	10	3.35		
Maximum Sharpe	9.92	3.3		
Minimum Volatility	0.54	0.59		

Portfolio of Bluechip Equity stock from US and India, Bond, Commodity, Bitcoin and Volatility index				
20.0% Target Volatility	75.89	20		
10.0% Target Return	10	2.24		
Maximum Sharpe	16.28	3.67		
Minimum Volatility	0.78	0.58		

Investors who had invested in wide range of assets like Equity, Debt, commodity including the new crypto currencies achieved 10% return with volatility of 3.35% under COVID19 (refer Exhibit 18). But by addition of VIX and IndiaVIX in the portfolio¹ the 10% return can be achieved with lesser volatility of 2.24%. Here also the performance of the portfolio of Equity, Debt, commodity, Cryptocurrency and Volatility asset class is almost same as portfolio of Equity, Debt and Volatility asset class in achieving 10% returns. This means the diversification benefits derived from gold and bitcoins can be produced by Volatility asset class. The efficient frontier of the portfolio is charted in Exhibit 17

Portfolio of Blue chip EQ stocks from US and India, US Debt ETF Commodity, Bitcoin and VIX ETF:

VIX index cannot be replicated by the underlying options since VIX index is the square root of the variance depicted by portfolio of out-of-the-money option chain with weight inversely proportional to square of option strikes. Hence to gain exposure to volatility index, one may have to invest in VIX derivatives or ETFs based on such derivatives. In developed markets, such ETFs are actively traded which can be used for investing in volatility asset class. Such ETFs still would not replicate the VIX performance as VIX represents expected volatility over next 30 days and one-month future on VIX would represent volatility over next 60 days. As volatility is mean reverting, the performance of both may not be aligned but by selecting ETFs (which focus on investing in short term VIX future) would serve the purpose of hedging the tail risk under extreme events, as it tends to be highly correlated. Hence to access the real-world portfolio performance, VIX and India VIX are replaced by with VIXY and VXX ETFs respectively as given below.

Exhibit 19A: Efficient Frontier of Equity stocks, Debt, Commodity ETF, Bitcoin and VIX ETF during peak of COVID19 (Apr 2019 to Mar 2020)

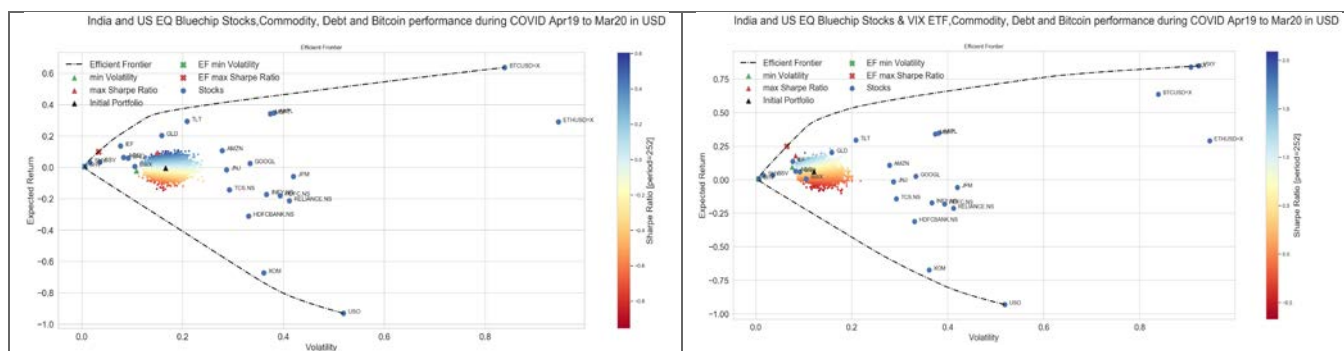


Exhibit 19B: Efficient Frontier of Equity stocks, Debt, Commodity ETF, Bitcoin and VIX ETF end of first COVID19 wave (Aug 2019 to Jul 2020)

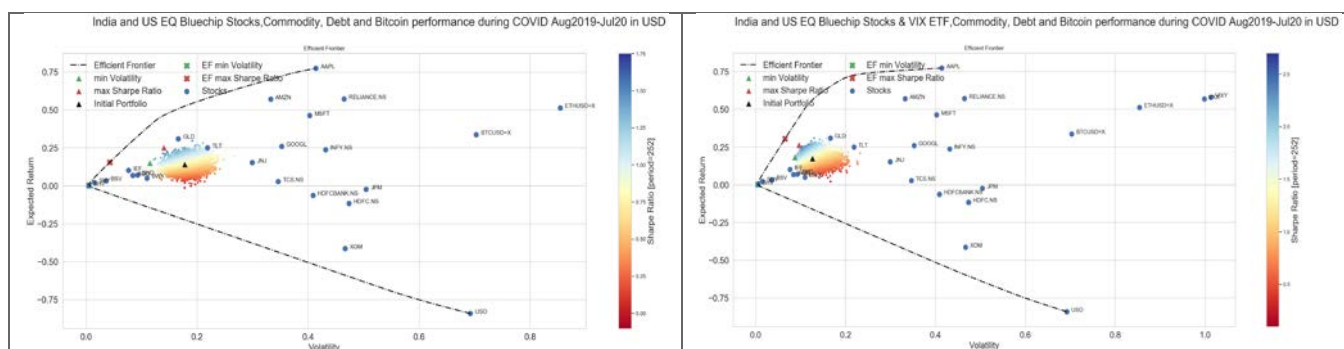


Exhibit 20: Portfolio analysis of Equity stock, Debt ETF, commodity ETF, Bitcoin and Volatility index ETF

Portfolio of Bluechip Equity stock from US and India, Bond, Commodity and Bitcoin				
Scenario	COVID 19 FY 2019-20		COVID 19 Aug 2019-July 20	
Performance Measure	Expected Return %	Volatility %	Expected Return %	Volatility %
20.0% Target Volatility	37.5	20	55.47	20
10.0% Target Return	10	3.35	10	2.74
Maximum Sharpe	9.92	3.3	15.53	4.27
Minimum Volatility	0.54	0.59	0.19	0.49
Portfolio of Bluechip Equity stock from US and India, Bond, Commodity, Bitcoin and VIX ETF				
20.0% Target Volatility	53.48	20	71.04	20
10.0% Target Return	10	2.54	10	2.16
Maximum Sharpe	24.88	6.43	30.54	6.54
Minimum Volatility	0.55	0.59	0.2	0.49

As only US VIX based ETF are available for trading which factors for the volatility of US markets but virus contagion effected global financial markets including USA. We observe from Exhibit 20 that by including VIX ETF (investable VIX) in the portfolio of equity, bond, commodity and Bitcoin was able to outperform the portfolio without VIX ETF during FY 2019-20. But the level of outperformance (expected return and volatility) was slightly reduced (Refer Exhibit 18) with VIX ETF than compared to VIX. Extending our analysis (rolling over few months) for another time frame i.e. August, 2019 to July 2020 (end of first

COVID19 wave) it was found (Exhibit 20) that adding VIX ETF was able to outperform the portfolio of equity, bond, commodity and Bitcoin during similar period. But the portfolio of equity, bond, commodity and Bitcoin during August 2019 to July 2020 provided superior performance than the above portfolio including VIX ETF during FY2019-20.

Robustness Checks:

Performance of Blue chip EQ stocks from US and India, US Debt ETF and Commodity during peak of tail event:

Exhibit 21A: Efficient Frontier of Equity stocks, Debt and Commodity ETF during peak of COVID19 pandemic, Mar2020

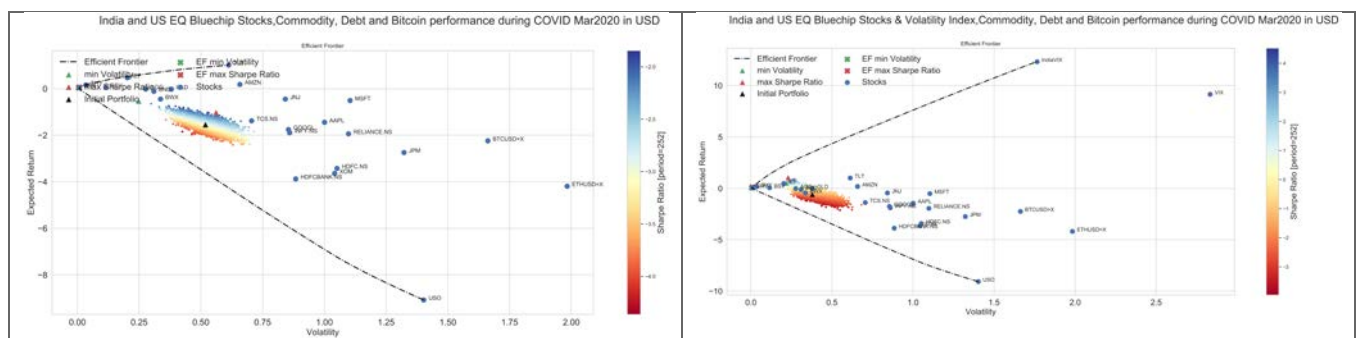


Exhibit 21B: Efficient Frontier of Equity stocks, Debt and Commodity ETF during peak of Financial Crisis, Sep-Dec2008

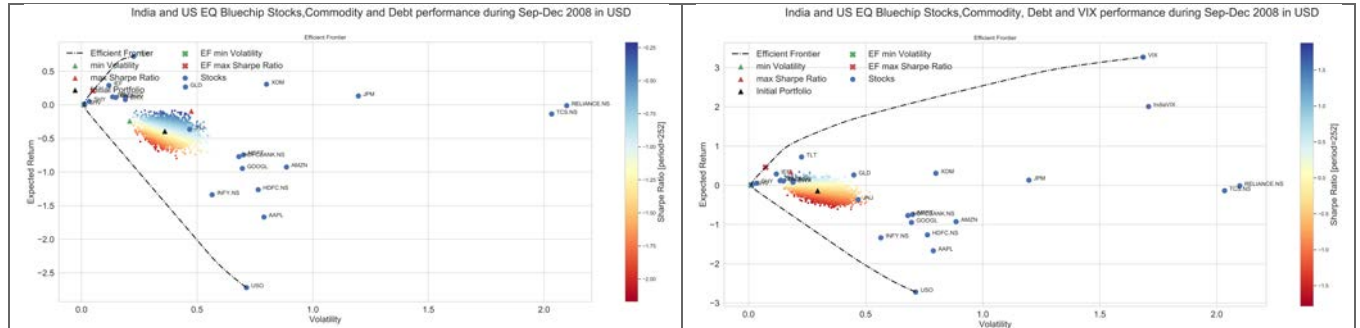


Exhibit 22: Portfolio analysis of Equity stock, Debt ETF, commodity ETF and Volatility index during peak time

Portfolio of Bluechip Equity stock from US and India, Bond ETF and Commodity				
Scenario	COVID 19 FY 2019-20		Financial Crisis FY 2008-09	
Performance Measure	Expected Return %	Volatility %	Expected Return %	Volatility %
20.0% Target Volatility	51.1	20	70.25	20
10.0% Target Return	10	1.57	10	2.38
Maximum Sharpe	5.7	0.65	20.99	4.95
Minimum Volatility	3.1	0.53	0.18	1.07
Portfolio of Bluechip Equity stock from US and India, Bond ETF, Commodity and Volatility index				
20.0% Target Volatility	216.24	20	110.78	20
10.0% Target Return	10	0.67	10	1.75

Maximum Sharpe	12.52	0.83	45.88	7.24
Minimum Volatility	4.58	0.48	0.62	1.06

Further, as robustness checks for the empirical results, the window of observation for portfolio formation and performance assessment was the period when the fat tail event happened i.e. September 2008 to December 2008 for Financial crises and March 2020 for COVID-19. The results (refer Exhibit 22) are quite consistent with our earlier observation while considering full financial year when the fat tail event happened. 10% return can be expected with volatility of 0.67% and 1.75% during peak of COVID19 and Financial Crisis in portfolio of Equity, Debt and commodity with addition of Volatility assets. Whereas without the volatility assets, the portfolio volatility observed was 1.57% and 2.38% respectively. The efficient frontier of the portfolio is charted in Exhibit 21.

Performance of Blue chip EQ stocks from US and India, US Debt ETF and Commodity in 2018-19 a year before COVID19:

As a next step, evaluating the performance of the portfolio one year before the outbreak of the COVID19 i.e. from Apr 2018 to Mar 2019 helped to access the suitability of asset class during low volatile regime. We observe (refer Exhibit 24) that adding the volatility asset didn't contribute in enhancing returns and same optimal return can be obtained without investment in VIX ETF (investable VIX). As volatility asset would play its part during extreme event and given inclusion of 3-6% proportion of VIX ETF had not led to significant reduction in return or increase in volatility during normal market regime indicates a good candidate for an ideal portfolio investment as well for passive investor.

Exhibit 23: Efficient Frontier of Equity stocks, Debt and Commodity ETF a year before the COVID19 pandemic, Apr18-Mar19

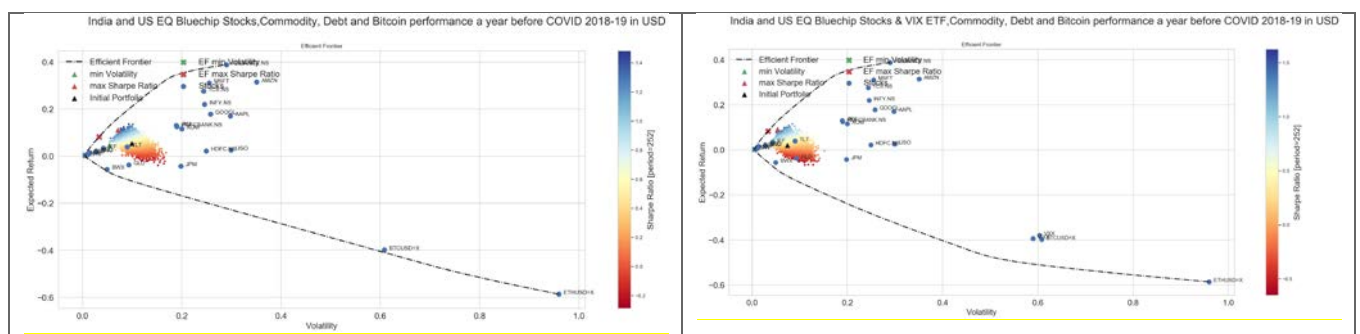


Exhibit 24: Portfolio analysis of Equity stock, Debt ETF, commodity ETF, Bitcoin and VIX ETF during 2018-19

Portfolio of Bluechip Equity stock from US and India, Bond ETF, Commodity and Bitcoin		
Scenario	FY 2018-19	

Performance Measure	Expected Return %	Volatility %		
20.0% Target Volatility	36.17	20		
10.0% Target Return	10	3.26		
Maximum Sharpe	12.62	4.15		
Minimum Volatility	0.4	0.53		
Portfolio of Bluechip Equity stock from US and India, Bond ETF, Commodity, Bitcoin and VIX ETF				
20.0% Target Volatility	35.26	20		
10.0% Target Return	10	4.06		
Maximum Sharpe	8.35	3.35		
Minimum Volatility	0.36	0.53		

From the portfolio performance analysis, we see the volatility asset class can meet the needs of investors with varying risk appetite in meeting their criteria of reduced portfolio volatility or increasing risk adjusted return. Investors having portfolio of investments across Equity, Debt, Commodity and Bitcoin asset classes can reduce their exposure to fewer asset classes i.e. Equity, Debt and Volatility to get the same level of portfolio performance. This would reduce the time and transaction cost involved in managing different asset classes along with ease of managing and tracking portfolio. Further, investors who are prepared for accepting higher volatility of 20% can maximize the expected return to 75% and 40% under the COVID19 and Financial crisis tail events respectively. The expected returns for investors who are willing to accept volatility of 20% further improved to 216% and 111% under the COVID19 and Financial crisis tail events respectively when the portfolios were constructed just before the events unfolded. Hence, at times of unfolding of Black swan event; portfolio of Equity, Debt and Volatility asset can provide not just protection from market moves but give superior expected returns and portfolio can be subsequently reallocated after the peak of the event. Another, interesting point we observed that the proportion of investment in volatility asset class (through Monte Carlo simulations) is around 5% to achieve minimum variance portfolio or target expected portfolio return of 10%.

Further in developed countries, ETFs based on VIX are available allowing investor to trade on volatility index. We checked the performance of ETF based on short term VIX futures performance similar to the underlying VIX under extreme events. But the portfolio with VIX based ETFs was not able to outperform the portfolio without VIX during normal market period i.e. 2018-19. Hence, for investor it is advisable to lower the exposure to volatility asset during normal market and increase the exposure when uncertainty builds in the market. Another good approach would be to track the VIX level and if VIX is below the long run

average it is better to increase the exposure in volatility product to 5%. Once the uncertainty had unfolded, one should cut down on the exposure in volatility product.

5. Summary and Conclusions

COVID19 pandemic event was quite uniquely characterised by frequent large jumps in asset prices within short span of time and market was not prepared for such kind of sudden shocks. Under such extreme tail events, most of the assets tend to be positively correlated breaking the diversification benefits. Large market shocks in short span of time give very little room for portfolio managers to churn their portfolio to reduce the down side risk. Tail risk hedging strategies come to the rescue of portfolio managers in these extreme market movements. The objective of this study is to explore “Volatility” as an asset class and explore its tail risk hedging performance under the two black swan effects i.e. COVID 19 and 2008 Financial Crisis. Firstly, we discuss the performance of different asset classes including Equity, FX, Bonds, Commodity, Bitcoins and VIX over these two tail risk events on a common scale. Subsequently, we construct efficient frontier portfolio optimization using various combinations of asset classes with and without Volatility Index to study the portfolio performance with respect to the two tail risk events. Empirical results show that by holding 5% or 1% of the portfolio allocation in volatility related asset class would have given portfolio insurance against the two Black Swan events. Moreover, the portfolio performance of Equity, Debt, Commodity and Bitcoins can be matched or outperformed with portfolio of Equity, Debt and Volatility asset class and with lesser number of stocks under each category.

Volatility is much more than a measure of level of uncertainty prevailing in the financial markets rather it has now become a new asset class for investors looking to diversify their portfolio strategy. All major derivatives markets have volatility indices but derivatives based on volatility are actively traded in US and Europe market. India has a volatility index, IndiaVIX based on Nifty option but no derivatives are available currently which reduces the potential of investing directly in India specific volatility asset. But ETF based on VIX and VSTOXX derivatives issued by Barclays, Proshares and Credit Suisse are actively traded which may be good option for investors for international diversification with portfolio insurance particularly for tail risk events. Recently mutual fund based on global index like S&P500 were launched in India and in similar fashion such funds can be launched with focus on volatility asset class till the domestic volatility asset develop. Trading on VIX needs good

expertise of fund managers to understand their risk exposures and take positions thereby providing sufficient liquidity in the markets. The contract specifications of the IndiaVIX derivative contracts are also important to catch its attraction amongst the traders and market participants.

The empirical results of the study have implications for investors, traders, policymakers and academia. Investors can look towards the effectiveness and importance of holding volatility as an asset class in avoiding market shortfalls by hedging downside risk particularly during Black Swan events. Treasuries of various corporate can use this asset class for effective risk management of their portfolio. Traders can make arbitrage profits by observing the VIX index and taking cues from the same to understand market mispricings (Implied vs Realized Volatility) in the short term if any in the derivatives and underlying segments. Policymakers can take signals about the risks being priced in markets and can use this information as surveillance for checking any excessive risk build-up in the financial system. Academia can benefit from the study to extend the scope of studying volatility in finance literature and consider Volatility as a measurable asset class for investors looking to diversify their portfolio strategy particularly at times of fat tail events.

Endnotes

1. Results of the Asset allocation in the portfolio by using Efficient Frontier analysis have been not presented due to brevity of space. The results can be provided by the authors on request.

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